B. Tech Degree VI Semester (Supplementary) Examination, October 2009

ME 603 MACHINE DESIGN I
(2006 Scheme)

Time : 3 Hours   Maximum Marks : 100

(Use of approved data book permitted)
(Missing data may be suitably assumed)

PART A
(Answer all questions)

(8 x 5 =40)

I.
a. Explain various steps in design process.
b. Explain how fatigue and creep are taken care of in design.
c. Explain what is meant by pre-loading of bolts. Why is it important in bolted assemblies?
d. Describe the various types of shaft couplings, mentioning the use of each type.
e. Discuss the failure modes in riveted joints.
f. Derive the relationship for deflection of a circular wire helical spring subjected to an axial load, W.
g. Discuss the advantages and disadvantages of welded joints in comparison to riveted joints.
h. Explain the term 'critical speed' of shaft.

PART B

(4 x 15 =60)

II. A long straight tube, 76 mm internal diameter and 2.5 mm thick is subjected to an internal pressure of 5-6 N/mm^2. Consider it as a thin cylinder. If the tube is subjected to a twisting moment of 70Nm, Elastic limit stress = $282 N/mm^2$. Determine the factors of safety on the three common theories of failure.

OR

III. A circular bar 500 mm. long is supported freely at its two ends. It is acted upon by a central concentrated cyclic load having a minimum value of 20 kN and a maximum value of 50 kN. Determine the diameter of the bar by taking a factor of safety of 1.5, size effect of 0.85, surface finish factor of 0.9. The material properties of bar are given by ultimate strength of 650 MPa, yield strength of 500 MPa and endurance strength of 350 Mpa.

IV. Acme threads are used in a lead screw of a lathe. Acme threads have 50mm outside diameter and 8 mm. pitch. The axial pressure required from the lead screw is 2500N. The collar subjected to thrust in the carriage has 110mm outside diameter and 55mm. inside diameter and the lead screw rotates at 30 r.p.m. Determine

(i) The power required to drive the lead screw
(ii) The efficiency of the lead screw

Assume μ for screw as 0.15 and that for collar as 0.12.

OR

(Turn over)

V. Design a muff coupling for transmitting 45 kw at a speed of 350 rev/min. The muff is made of cast iron and is used to connect two shafts. The allowable shear stress in the material of the shaft is $50N/mm^2$. The material of the key and shaft is same and the coupling is required to transmit 25% overload. Allowable shear stress for cast iron is $16N/mm^2$.

VI. Find the efficiency of the following joints:

(i) Single riveted lap joint of 6 mm plates with 20 mm diameter rivets having a pitch of 50mm.

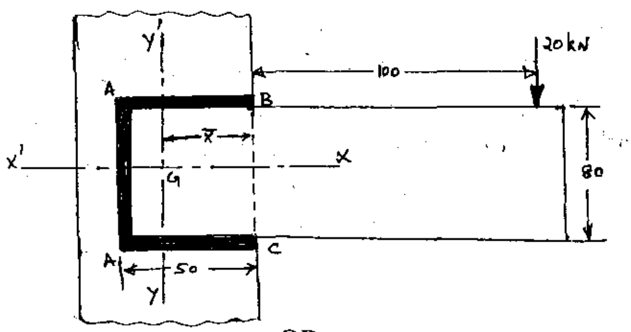
(ii) Double rivetted lap joint of 6mm plates with 20mm diameter rivets having a pitch of 60mm.

Permissible tensile stress in plate	= 100 MPa
Permissible shearing stress in rivets	= 80 MPa
Permissible crushing stress in rivets	= 160 MPa

OR

VII. A cylinder relief valve of diameter 60mm. is working at a pressure of $1.5N/mm^2$ and lifts 6mm for 5% increase in pressure. Design the valve spring by computing important parameters of the spring. The allowable shear stress is $500N/mm^2$ and $G = 0.85 \times 10^5 N/mm^2$ for the spring material.

VIII. The weld shown in figure below is subjected to a vertical load of 20kN. The throat of the fillet weld is 8mm. Calculate the maximum shear stress in the weld.



OR

IX. A shaft supported at the ends in ball bearings carries a straight tooth spur gear at its mid span and is to transmit 7.5 kW at 280 r.p.m. The pitch circle diameter of the gear is 150 mm. The distance between the centre line of bearings and gear are 100 mm. each. If the shaft is made of steel and the allowable shear stress is 40MPa, determine the diameter of the shaft. Show in a sketch how the gear will be mounted on the shaft; also indicate the ends where the bearings will be mounted. The pressure angle of the gear may be taken as 20^0.

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